

Notes on Fisher matrix approximation

This note recapitulates the following approximations of the Empirical Fisher Matrix :
KFAC, EKFAC, TKFAC, TEKFAC, Adafisher.

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Chapter 1

Fisher matrix

1.1 Overview and main findings

Adafisher aims to replace Adam's second-moment with a Kronecker-factored

1.2 Neural networks as a probabilistic model

1.3 The Fisher information matrix, the Hessian, and the Gauss–Newton matrix

1.4 Natural gradient descent

1.5 The empirical Fisher : definition and limitations

1.6 Kronecker products and the rvec/cvec convention

Chapter 2

Approximation of the Empirical Fisher

2.1 KFAC : a block-diagonal approximation

2.1.1 The Kronecker-factored approximation to a layer's Fisher block

2.1.2 Independence assumptions

2.1.3 Structured inverses and damping

2.1.4 Estimating the factors, and momentum

2.1.5 Extension to convolutional layers

2.2 EKFac : an eigenvalue-corrected Kronecker factorization

2.2.1 Motivation : sub-optimality of KFAC’s implicit diagonal

2.2.2 The Kronecker-factored eigenbasis

2.2.3 Optimal diagonal rescaling in the eigenbasis

2.2.4 Estimating the corrected eigenvalues

2.3 TKFAC : a trace-restricted Kronecker-factored approximation

2.3.1 The quadratic form estimator

2.3.2 TKFAC for fully-connected layers

2.3.3 TKFAC for convolutional layers

2.3.4 Approximation error and trace preservation

2.4 TEKFAC : eigenvalue correction of TKFAC

2.4.1 Construction

2.4.2 Damping and the exponential moving averages

2.4.3 Discussion : relation to KFAC, EKFAC and TKFAC

2.5 Adafisher : a diagonal Fisher approximation

2.5.1 Diagonal concentration of the Kronecker factors

2.5.2 Efficient computation of the Fisher information matrix

2.5.3 Min-max normalization and the exponential moving average

2.5.4 Integration into an adaptive optimization framework

2.5.5 Convergence analysis

Chapter 3

Unified view

3.1 A common template : the Kronecker-factored second moment

3.2 Comparison of the five rescalings

3.3 Notation conventions used throughout these notes

Bibliography

- Felix Dangel, Bálint Mucsányi, Tobias Weber, and Runa Eschenhagen. Kronecker-factored approximate curvature (KFAC) from scratch, 2025.
- Kai-Xin Gao, Xiao-Lei Liu, Zheng-Hai Huang, Min Wang, Shuangling Wang, Zidong Wang, Dachuan Xu, and Fan Yu. Eigenvalue-corrected natural gradient based on a new approximation, 2020a.
- Kai-Xin Gao, Xiao-Lei Liu, Zheng-Hai Huang, Min Wang, Zidong Wang, Dachuan Xu, and Fan Yu. A trace-restricted Kronecker-factored approximation to natural gradient, 2020b.
- Thomas George, César Laurent, Xavier Bouthillier, Nicolas Ballas, and Pascal Vincent. Fast approximate natural gradient descent in a Kronecker-factored eigenbasis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Roger Grosse and James Martens. A Kronecker-factored approximate fisher matrix for convolution layers, 2016.
- Frederik Kunstner, Lukas Balles, and Philipp Hennig. Limitations of the empirical fisher approximation for natural gradient descent, 2019.
- James Martens and Roger Grosse. Optimizing neural networks with Kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015.
- Damien Martins Gomes, Yanlei Zhang, Eugene Belilovsky, Guy Wolf, and Mahdi S. Hosseini. AdaFisher: Adaptive second order optimization via fisher information. In *International Conference on Learning Representations (ICLR)*, 2025.